

My Design: Adaptive Technology
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Element E: Application of STEM Principles and Practices

All of our work with STEM principles so far have branched off of our measurements of the middle island and cubbies in the sensory room. Those measurements are seen below:

<u>Middle Island:</u> - Width of middle island from the outside: 12 $\frac{7}{8}$ inches - Width of middle island from inside: 10 $\frac{7}{8}$ inches - Length of half the middle island: 34 $\frac{1}{2}$ inches - Length of island end to end from outside: 72 inches - Height from bottom to lip: 44 inches - Height of island from outside: 42 $\frac{1}{2}$ inches - Height of island from inside (this should be height of door): 40 $\frac{9}{16}$ inches - Width from birds eye view of island: 11 $\frac{15}{16}$ inches - Length of middle divider, pillars, and lips: 1 $\frac{1}{16}$ inches - Height off the ground or of base: 4 $\frac{5}{8}$ inches - Depth of base//stand: 1 $\frac{11}{16}$ inches - Shelf stand length: 1 $\frac{1}{14}$ inch - Height of shelf stand: 1 $\frac{5}{8}$ inch - Shelf length: 34 inches - Shelf width: 10 inches - Shelf height: 1 $\frac{3}{4}$ inches	<u>Cubby:</u> - Length of both side cubbies wall to end from outside: 73 $\frac{1}{2}$ inches - Height cubby above ground: 3 $\frac{1}{2}$ inches - Depth out from base stand: 2 $\frac{7}{8}$ inches - Height end to end from outside: 42 $\frac{1}{2}$ inches - Height inside: 40 $\frac{3}{4}$ inches - Depth of cubby from inside: 11 $\frac{1}{16}$ inches - Inside length of one cubby from inside: 35 $\frac{1}{4}$ inches - Depth from top of cubby: 11 $\frac{3}{8}$ inches - Total height of side white shelves from top of cubbies: 36 $\frac{1}{2}$ inches - Length of white shelves: 71 $\frac{7}{8}$ inches - Width of white shelves: 12 $\frac{1}{2}$ inches
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By creating doors that can comply with the middle island's shape we can develop more storage space that makes use of unoptimized space. The qualities we focused on for the design was safety, adaptability, and optimization of storage. The middle island worked

functionally as just a large block between the sensory space and the rest of the classroom, but by adding coverage to it we can create a new place for storing items and help define the area as a focused sensory space. The children's needs are largely centered in the safety aspect of the design. Ensuring the doors are sturdy will help protect the kids from potentially hurting themselves by putting too much force on the doors and causing them to collapse. Throughout the design of the area they try to avoid sharp edges and so rounding out edges can help.

When looking to determine the weight capacity of the hinges we tried to calculate the center of mass of the boards. We have three hinges between sides of a foldable door, meaning we could use center of mass to calculate where to position the top and bottom hinges since we know that one hinge will be in the center.

To start, we needed to calculate the mass of one of our doors, which covers half of the middle island. Through research, we found the density of the middle island which we used with the volume of our door to find the weight of our door. When using this mass for the center of mass, it ended up canceling out but it was still helpful to find the mass of our door in case we need it in the future, After finding the mass, we moved on to calculating the center of mass. Those calculations along with the mass calculations are shown below:

Density of Birch wood = 42 lbs/ft^3

Mass of one door (1/2 island) = $\frac{42 \text{ lbs}}{\text{ft}^3} \times \frac{(\text{ft})^3}{(12 \text{ in})^3} \times \frac{3 \text{ ft} \times 40.5 \text{ in} \times 34.5 \text{ in}}{12 \text{ in}} = 25.47 \text{ lbs}$

Mass hinge = 0.1 kg Mass of section door (1/3 door) = $\frac{25.47}{3} = 8.49 \text{ lbs} = 3.85 \text{ kg}$

Diagram of a door with hinges at 3 in, 20.25 in, and 40.5 in from the top. The center of mass is calculated as $X_{cm} = 20.25 \text{ in}$.

Formulas used:

$$X_{cm} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3}$$

Since $m_1 = m_2 = m_3$ (mass is evenly distributed),

$$X_{cm} = \frac{m(x_1 + x_2 + x_3)}{3m}$$

$$X_{cm} = \frac{(x_1 + x_2 + x_3)}{3}$$

Testing Potential heights of hinges:

$$X_{cm} = \frac{3 \text{ in} + 20.25 \text{ in} + (40.5 \text{ in} - 3 \text{ in})}{3} = 60.75/3$$

$$X_{cm} = 20.25 \text{ in}$$

Conclusion: We calculated the center of mass using hinge placements at the center, 3 inches from the top, and 3 inches from the bottom and got 20.25 inches, which matches the halfway point of the door, which is the overall center of mass. This conceptually makes sense because we are dividing the mass of the door equally across three hinges, which means that in our center of mass formula $m_1 = m_2 = m_3$. Because of this, we have to divide the positions of the three hinges (x_1 , x_2 , and x_3) by the number of masses added (which is 3), which matches up with the true center of mass which is at the center of the door because the mass is evenly distributed.

Volume of Middle Island (from inside) = $L \times W \times H$

$$V_m = 5.75 \text{ ft} \times 0.9063 \text{ ft} \times 3.380 \text{ ft} = 17.614 \text{ ft}^3$$

$$V_m = 17.614 \text{ ft}^3$$

Surface Area of Middle Island = $2LW + 2LH + 2WH$

$$SA_m = 2(5.75)(0.9063) + 2(5.75)(3.38) + 2(0.9063)(3.38) = 55.419 \text{ ft}^2$$

$$SA_m = \mathbf{55.419 \text{ ft}^2}$$

Volume of Cubby (from inside) = $L \times W \times H$

$$V_c = 5.875 \text{ ft} \times 0.92 \text{ ft} \times 3.40 \text{ ft} = 18.377 \text{ ft}^3$$

$$V_c = \mathbf{18.377 \text{ ft}^3}$$

Surface Area of Cubby = $2LW + 2LH + 2WH$

$$SA_c = 2(5.875)(0.92) + 2(0.92)(3.4) + 2(3.4)(5.875) = 57.016 \text{ ft}^2$$

$$SA_c = \mathbf{57.016 \text{ ft}^2}$$

Because we are storing items in the middle island and cubbies, we figured it would be helpful to calculate the volume of both spaces so we could get an idea of capacity storage. It is also useful to know the surface area, in case we need that calculation later down the line.

Expert Review:

Nathan Johnson is a vehicle engineering associate at *Exponent*, a global engineering and scientific consulting firm. He has experience with post car collision analysis, and other various components of engineering, math, and science. He has noted that the sliding and pocket doors would be best because they remove hazards that a regular door would bring to the area. He also recommended that we not overcomplicate things, and split the project up into appropriate components that would allow us to work on one at a time.

Doctor Joseph Sarver is a professor at Drexel University School of Biomedical Engineering, Science and Health Systems. He has a B.S. Mechanical Engineering from Purdue University, an M.S. Biomedical Engineering from Drexel University, a PH.D. Biomedical Engineering from Drexel University, and did his Post-Doc Department of Orthopaedic Surgery at University of Pennsylvania. He approved of our designs including the design that we want to go with after meeting with the stakeholders which were the pocket door and the bi-fold door design that reduces danger when it's opened and the ball and chain cut-out to place tactiles onto.